

# Assessment Programs

Questions 1 to 5 — Program Output (IDLE Shell)

## Question 1: Water Jug Problem — Goal-Based Agent

```
Enter 4-gallon jug capacity: 4
Enter 3-gallon jug capacity: 3
4-Gallon Jug: 2
3-Gallon Jug: 0
```

## Question 2: Smart Farm Agent — Model-Based Goal-Oriented Agent

```
Enter number of percepts: 3
Enter percept: CAMERA
Enter percept: TEMPERTURE
Enter percept: SOIL DATA
Percept: CAMERA
Percept: TEMPERTURE
Percept: SOIL DATA
Goal: Monitor Farm Conditions
Agent Type: Model-Based Goal-Oriented Agent
```

## Question 3: 8-Queens Problem — Backtracking Search

```
Q . . . . .
. . . . . Q .
. . . . Q . .
. . . . . . Q
. Q . . . . .
. . . Q . . .
. . . . . Q .
. . Q . . . .
```

## Question 4: Cab Booking Agent

```
Enter number of cab types: 4
Enter cab type: MINI
Enter cab type: MICRO
Enter cab type: SEDAN
Enter cab type: PRIME
```

```
Available Cabs: ['MINI', 'MICRO', 'SEDAN', 'PRIME']
Enter cab type to book: SEDAN
Booking Confirmed: SEDAN
```

## Question 5: Uniform Cost Search (UCS) Agent

Enter Start Node: S

Enter Goal Node: G

Agent Type: Goal-Based AI Agent

Path: ['S', 'A', 'C', 'G']

Cost: 4